

# Phase 1 - Segmentation

Segmentation used fixed settings on every image: percentile normalise (p1-p99.5), gaussian sigma 1.5, lower multi-Otsu for pore, multi-Otsu for silicon (the bright tail has no separate histogram mode - the valley search falls back to multi-Otsu on every image), opening + watershed. Pixel size 25.0 nm/px from TIFF resolution (FOV ~175x54 um). Detectors are pixel-aligned (zero-shift cross-correlation), so Inlens-dark pixels next to pores refine cracks. Median pore fraction shifts <~10% at +/-10% threshold; left/right pore fraction correlation 0.87; two low-contrast images are flagged by the quality guards.

Assumption (applies throughout): the grey bulk is assumed graphite, bright particles are assumed silicon-based, black is pore/crack. This comes from image appearance and is unconfirmed.

One method, one set of settings for all 31 images. Classes: pore/crack (black), graphite bulk (grey), silicon particle (bright). No binder class was invented: intensity between bulk and pore did not form a separable third phase.

| batch   | image_id | um_per_px | pore_frac | silicon_frac | t_lo   | t_bright |
|---------|----------|-----------|-----------|--------------|--------|----------|
| Batch_1 | 4ih2ggld | 0.025     | 0.1134    | 0.1686       | 0.2819 | 0.5697   |
| Batch_1 | 5n1q8atc | 0.025     | 0.08157   | 0.184        | 0.289  | 0.5722   |
| Batch_1 | f1vzngrs | 0.025     | 0.08005   | 0.08262      | 0.2456 | 0.5744   |
| Batch_1 | ffwubibz | 0.025     | 0.09383   | 0.04862      | 0.29   | 0.6357   |
| Batch_1 | fzrt2k6r | 0.025     | 0.1061    | 0.06837      | 0.3005 | 0.6252   |
| Batch_1 | iv6g2oq0 | 0.025     | 0.1116    | 0.06101      | 0.2523 | 0.6027   |
| Batch_1 | uhdslk0o | 0.025     | 0.1044    | 0.0651       | 0.2804 | 0.6155   |
| Batch_2 | 3806gxp0 | 0.025     | 0.1208    | 0.06052      | 0.2521 | 0.5908   |
| Batch_2 | avn74qx1 | 0.025     | 0.1127    | 0.04463      | 0.2466 | 0.6061   |
| Batch_2 | b3esycq1 | 0.025     | 0.1079    | 0.08769      | 0.2994 | 0.6161   |
| Batch_2 | epqdaau9 | 0.025     | 0.09437   | 0.06753      | 0.2285 | 0.5527   |
| Batch_2 | i9jiqjwl | 0.025     | 0.1012    | 0.07411      | 0.2735 | 0.5916   |
| Batch_2 | r17byphk | 0.025     | 0.1246    | 0.05361      | 0.285  | 0.6145   |
| Batch_2 | rxax5ozo | 0.025     | 0.0996    | 0.05794      | 0.2756 | 0.6109   |
| Batch_3 | 0grcilhi | 0.025     | 0.1511    | 0.06531      | 0.282  | 0.5963   |
| Batch_3 | 71vgq3fw | 0.025     | 0.0944    | 0.07265      | 0.2702 | 0.5922   |
| Batch_3 | 9luzk4jm | 0.025     | 0.1499    | 0.0611       | 0.3087 | 0.6499   |
| Batch_3 | cfe5vt7s | 0.025     | 0.1078    | 0.07043      | 0.2772 | 0.6068   |
| Batch_3 | hawkfj64 | 0.025     | 0.1046    | 0.04905      | 0.2793 | 0.6074   |
| Batch_3 | hzumfsms | 0.025     | 0.1638    | 0.05008      | 0.3197 | 0.6606   |
| Batch_3 | kbdh4tri | 0.025     | 0.09401   | 0.08897      | 0.2676 | 0.5644   |
| Batch_3 | mgxahqnk | 0.025     | 0.108     | 0.07021      | 0.2826 | 0.614    |
| Batch_3 | pl8uabbv | 0.025     | 0.1168    | 0.06048      | 0.2603 | 0.6024   |
| Batch_3 | ptg8lmto | 0.025     | 0.09295   | 0.05595      | 0.27   | 0.6002   |
| Batch_3 | tuy3zymq | 0.025     | 0.09924   | 0.09294      | 0.2676 | 0.5564   |
| Batch_3 | ufdvpb81 | 0.025     | 0.1291    | 0.05734      | 0.3218 | 0.6533   |

| batch   | image_id | um_per_px | pore_frac | silicon_frac | t_lo   | t_bright |
|---------|----------|-----------|-----------|--------------|--------|----------|
| Batch_3 | utfgcjfa | 0.025     | 0.1112    | 0.05933      | 0.268  | 0.6081   |
| Batch_3 | vc2whyag | 0.025     | 0.1101    | 0.05998      | 0.2739 | 0.6102   |
| Batch_3 | x77cy643 | 0.025     | 0.1101    | 0.06491      | 0.2719 | 0.6113   |
| Batch_3 | x7u69zsw | 0.025     | 0.09187   | 0.09659      | 0.2598 | 0.5723   |
| Batch_3 | xgj4xftb | 0.025     | 0.1072    | 0.05959      | 0.2732 | 0.5893   |

**Threshold sensitivity: relative change in each phase fraction when each threshold is shifted +/-10% (median over images).**

| param        | shift | phase    | rel_change |
|--------------|-------|----------|------------|
| t_b(silicon) | -0.1  | pore     | -0.0008269 |
| t_b(silicon) | -0.1  | silicon  | 0.3036     |
| t_b(silicon) | -0.1  | graphite | -0.02669   |
| t_b(silicon) | 0     | pore     | 0          |
| t_b(silicon) | 0     | silicon  | 0          |
| t_b(silicon) | 0     | graphite | 0          |
| t_b(silicon) | 0.1   | pore     | 3.751e-05  |
| t_b(silicon) | 0.1   | silicon  | -0.1346    |
| t_b(silicon) | 0.1   | graphite | 0.01187    |
| t_lo(pore)   | -0.1  | pore     | -0.1163    |
| t_lo(pore)   | -0.1  | silicon  | 0          |
| t_lo(pore)   | -0.1  | graphite | 0.01429    |
| t_lo(pore)   | 0     | pore     | 0          |
| t_lo(pore)   | 0     | silicon  | 0          |
| t_lo(pore)   | 0     | graphite | 0          |
| t_lo(pore)   | 0.1   | pore     | 0.1308     |
| t_lo(pore)   | 0.1   | silicon  | 0          |
| t_lo(pore)   | 0.1   | graphite | -0.01606   |

**Left/right half agreement (independent segmentation of each half) and cross-detector agreement.**

| batch   | image_id | pore_L  | pore_R  | silicon_L | silicon_R |
|---------|----------|---------|---------|-----------|-----------|
| Batch_1 | 4ih2ggld | 0.1025  | 0.1233  | 0.1775    | 0.1603    |
| Batch_1 | 5n1q8atc | 0.08753 | 0.07531 | 0.164     | 0.2082    |
| Batch_1 | f1vzngrs | 0.07401 | 0.08412 | 0.09518   | 0.07152   |
| Batch_1 | ffwubibz | 0.09539 | 0.09105 | 0.04913   | 0.04826   |
| Batch_1 | fzrt2k6r | 0.1023  | 0.1096  | 0.06006   | 0.07687   |
| Batch_1 | iv6g2oq0 | 0.1124  | 0.1121  | 0.04253   | 0.07959   |

| batch   | image_id | pore_L  | pore_R  | silicon_L | silicon_R |
|---------|----------|---------|---------|-----------|-----------|
| Batch_1 | uhdslk0o | 0.1048  | 0.1049  | 0.06575   | 0.06431   |
| Batch_2 | 3806gxp0 | 0.1119  | 0.1271  | 0.07077   | 0.05029   |
| Batch_2 | avn74qx1 | 0.1133  | 0.1107  | 0.03346   | 0.05568   |
| Batch_2 | b3esycq1 | 0.1137  | 0.1037  | 0.09131   | 0.08409   |
| Batch_2 | epqdaau9 | 0.09268 | 0.09594 | 0.08949   | 0.04407   |
| Batch_2 | i9jiqjwl | 0.09354 | 0.11    | 0.0678    | 0.0817    |
| Batch_2 | r17byphk | 0.1248  | 0.1207  | 0.05937   | 0.0566    |
| Batch_2 | rxax5ozo | 0.09236 | 0.1059  | 0.06826   | 0.04815   |
| Batch_3 | 0grcilhi | 0.1514  | 0.1476  | 0.08277   | 0.05179   |
| Batch_3 | 71vgq3fw | 0.09699 | 0.09252 | 0.08482   | 0.06075   |
| Batch_3 | 9luzk4jm | 0.1403  | 0.1521  | 0.06409   | 0.05801   |
| Batch_3 | cfe5vt7s | 0.1114  | 0.1061  | 0.07529   | 0.06756   |
| Batch_3 | hawkfj64 | 0.1026  | 0.1084  | 0.04892   | 0.05295   |
| Batch_3 | hzumfsms | 0.1712  | 0.1577  | 0.05186   | 0.04829   |
| Batch_3 | kbdh4tri | 0.09256 | 0.09814 | 0.08721   | 0.09113   |
| Batch_3 | mgxahqnk | 0.0981  | 0.1169  | 0.04286   | 0.0973    |
| Batch_3 | pl8uabbv | 0.1258  | 0.1098  | 0.06004   | 0.06187   |
| Batch_3 | ptg8lmto | 0.08997 | 0.0973  | 0.06917   | 0.04274   |
| Batch_3 | tuy3zymq | 0.09492 | 0.1043  | 0.1065    | 0.08092   |
| Batch_3 | ufdvpb81 | 0.1254  | 0.1344  | 0.03723   | 0.07804   |
| Batch_3 | utfgcjfa | 0.1114  | 0.1127  | 0.05626   | 0.06229   |
| Batch_3 | vc2whyay | 0.109   | 0.1103  | 0.06661   | 0.05309   |
| Batch_3 | x77cy643 | 0.111   | 0.109   | 0.08096   | 0.04904   |
| Batch_3 | x7u69zsw | 0.0834  | 0.1015  | 0.09888   | 0.09343   |
| Batch_3 | xgj4xftb | 0.1069  | 0.1065  | 0.06174   | 0.06491   |

| iou_pore | dice_pore | pore_frac_inlens | batch   | image_id |
|----------|-----------|------------------|---------|----------|
| 0.145    | 0.2532    | 0.7073           | Batch_1 | 4ih2ggld |
| 0.09984  | 0.1815    | 0.7426           | Batch_1 | 5n1q8atc |
| 0.1045   | 0.1893    | 0.7116           | Batch_1 | f1vzngrs |
| 0.1434   | 0.2508    | 0.5475           | Batch_1 | ffwubibz |
| 0.3296   | 0.4957    | 0.2646           | Batch_1 | fzrt2k6r |
| 0.147    | 0.2563    | 0.6576           | Batch_1 | iv6g2oq0 |
| 0.1316   | 0.2325    | 0.693            | Batch_1 | uhdslk0o |
| 0.1395   | 0.2448    | 0.6681           | Batch_2 | 3806gxp0 |
| 0.1354   | 0.2386    | 0.6831           | Batch_2 | avn74qx1 |
| 0.2911   | 0.451     | 0.2927           | Batch_2 | b3esycq1 |

| iou_pore | dice_pore | pore_frac_inlens | batch   | image_id |
|----------|-----------|------------------|---------|----------|
| 0.1218   | 0.2171    | 0.7092           | Batch_2 | epqdaau9 |
| 0.1307   | 0.2312    | 0.7262           | Batch_2 | i9jiqjwl |
| 0.1846   | 0.3117    | 0.53             | Batch_2 | r17byphk |
| 0.125    | 0.2222    | 0.676            | Batch_2 | rxax5ozo |
| 0.1902   | 0.3196    | 0.6413           | Batch_3 | 0grcilhi |
| 0.104    | 0.1885    | 0.6839           | Batch_3 | 71vgq3fw |
| 0.1735   | 0.2957    | 0.7744           | Batch_3 | 9luzk4jm |
| 0.146    | 0.2548    | 0.6182           | Batch_3 | cfe5vt7s |
| 0.1167   | 0.209     | 0.6708           | Batch_3 | hawkfj64 |
| 0.2548   | 0.4061    | 0.5313           | Batch_3 | hzumfsms |
| 0.1029   | 0.1865    | 0.6781           | Batch_3 | kbdh4tri |
| 0.1252   | 0.2226    | 0.6369           | Batch_3 | mgxahqnk |
| 0.149    | 0.2594    | 0.7079           | Batch_3 | pl8uabbv |
| 0.1017   | 0.1846    | 0.7295           | Batch_3 | ptg8lmt0 |
| 0.1091   | 0.1968    | 0.687            | Batch_3 | tuy3zymq |
| 0.2482   | 0.3977    | 0.4164           | Batch_3 | ufdvpb81 |
| 0.1439   | 0.2516    | 0.6527           | Batch_3 | utfgcjfa |
| 0.1388   | 0.2437    | 0.6743           | Batch_3 | vc2whyay |
| 0.1462   | 0.2551    | 0.6157           | Batch_3 | x77cy643 |
| 0.09617  | 0.1755    | 0.6472           | Batch_3 | x7u69zsw |
| 0.1093   | 0.1971    | 0.6527           | Batch_3 | xgj4xftb |

Cross-detector note: raw pixel IoU between an Inlens-only dark mask and the BSE pore mask is low (~0.1-0.2) because the Inlens detector is texturally dark over most of the surface - fine surface detail is not the same as void space. Inlens is therefore only used to refine pixels within 2 px of a BSE pore, not as a standalone pore detector.

### ***Image-quality guards; flagged images are excluded in parallel analyses:***

| batch   | image_id | si_bulk_contrast | noise_mad | sharpness | shading_range | flagged | flag_reason                                |
|---------|----------|------------------|-----------|-----------|---------------|---------|--------------------------------------------|
| Batch_1 | 4ih2ggld | 2.149            | 11.86     | 11.22     | 43.91         | True    | low Si/bulk contrast; high noise; high noi |
| Batch_1 | 5n1q8atc | 2.269            | 11.86     | 11.42     | 42.14         | True    | low Si/bulk contrast; high noise; high noi |
| Batch_1 | f1vzngrs | 4.053            | 10.38     | 10.55     | 43.2          | False   | nan                                        |
| Batch_1 | ffwubibz | 4.097            | 10.38     | 9.86      | 36.62         | False   | nan                                        |
| Batch_1 | fzrt2k6r | 3.625            | 10.38     | 10.29     | 40.52         | False   | nan                                        |
| Batch_1 | iv6g2oq0 | 4.089            | 10.38     | 10.13     | 47.6          | False   | nan                                        |
| Batch_1 | uhdsk0o  | 3.845            | 10.38     | 10.39     | 43.72         | False   | nan                                        |
| Batch_2 | 3806gxp0 | 4.278            | 10.38     | 9.841     | 41.19         | False   | nan                                        |
| Batch_2 | avn74qx1 | 4.408            | 10.38     | 9.71      | 41.55         | False   | nan                                        |
| Batch_2 | b3esycq1 | 3.617            | 10.38     | 10.37     | 43.48         | False   | nan                                        |
| Batch_2 | epqdaau9 | 4.447            | 10.38     | 10.52     | 46.87         | False   | nan                                        |

| batch   | image_id | si_bulk_contrast | noise_mad | sharpness | shading_range | flagged | flag_reason |
|---------|----------|------------------|-----------|-----------|---------------|---------|-------------|
| Batch_2 | i9jiqjwl | 3.643            | 10.38     | 10.29     | 38.16         | False   | nan         |
| Batch_2 | r17byphk | 3.66             | 10.38     | 9.821     | 44.49         | False   | nan         |
| Batch_2 | rxax5ozo | 4.297            | 10.38     | 9.537     | 41.12         | False   | nan         |
| Batch_3 | 0grcilhi | 3.78             | 8.896     | 9.083     | 50.2          | False   | nan         |
| Batch_3 | 71vgq3fw | 4.141            | 7.413     | 6.954     | 32.67         | False   | nan         |
| Batch_3 | 9luzk4jm | 4.634            | 7.413     | 7.314     | 36.82         | False   | nan         |
| Batch_3 | cfe5vt7s | 3.946            | 10.38     | 9.992     | 41.84         | False   | nan         |
| Batch_3 | hawkfj64 | 4.345            | 8.896     | 9.16      | 36.16         | False   | nan         |
| Batch_3 | hzumfsms | 4.619            | 7.413     | 7.202     | 48.94         | False   | nan         |
| Batch_3 | kbdh4tri | 3.7              | 7.413     | 7.02      | 26.3          | False   | nan         |
| Batch_3 | mgxahqnk | 4.397            | 8.896     | 9.277     | 46.74         | False   | nan         |
| Batch_3 | pl8uabbv | 3.983            | 10.38     | 10.15     | 42.57         | False   | nan         |
| Batch_3 | ptg8lmt0 | 4.398            | 8.896     | 9.702     | 39.92         | False   | nan         |
| Batch_3 | tuy3zymq | 3.403            | 7.413     | 6.953     | 27.7          | False   | nan         |
| Batch_3 | ufdvpb81 | 4.506            | 7.413     | 7.268     | 36.82         | False   | nan         |
| Batch_3 | utfgcjfa | 4.495            | 10.38     | 9.466     | 47.68         | False   | nan         |
| Batch_3 | vc2whyay | 4.487            | 10.38     | 9.503     | 44.29         | False   | nan         |
| Batch_3 | x77cy643 | 4.437            | 10.38     | 9.546     | 38.76         | False   | nan         |
| Batch_3 | x7u69zsw | 3.974            | 7.413     | 7.095     | 34.53         | False   | nan         |
| Batch_3 | xgj4xftb | 4.1              | 8.896     | 9.517     | 46.6          | False   | nan         |

**Example overlays (grey=graphite, orange=silicon, blue=pore) and per-batch contact sheets:**



